

Hot Asphalt Mix (HMA) Material Specifications:

Must conform to the Uniform Guide Specifications 32.12.15.13

No Reclaimed Asphalt / Type B / Binder: 67 -22

Aggregates:

- Shall consist of crushed stone, crushed gravel, crushed slag, screenings, natural sand and mineral filler, as required. The portion of material retained on the 4.75 mm No. 4 sieve is coarse aggregate. The portion of material passing the 4.75 mm No. 4 sieve and retained on the 0.075 mm No. 200 sieve is fine aggregate. The portion passing the 0.075 mm No. 200 sieve is defined as mineral filler. Aggregate testing shall have been performed within 90 days of performing the mix design.

Coarse Aggregate:

- Percentage of Wear (ASTM C131) must not exceed 40. Aggregates with a higher percentage of wear may be specified, provided a satisfactory record under similar conditions of service and exposure has been demonstrated.
- Coarse aggregate shall consist of sound, tough, durable particles, free from films of material that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances.
- The sodium sulfate soundness loss shall not exceed 12 percent, or the magnesium sulfate soundness loss shall not exceed 18 percent after five cycles when tested in accordance with ASTM C88.

Fine Aggregate:

- Fine aggregate shall consist of clean, sound, tough, durable particles. The aggregate particles shall be free from coatings of clay, silt, or any objectionable material and shall contain no clay balls. The fine aggregate particles shall meet the following requirements:

Mineral Filler:

- Mineral filler shall be nonplastic material meeting the requirements of ASTM D242/D242M.

Aggregate Gradation: The combined aggregate gradation shall conform to table 7:

Table 7 – Aggregate Gradation 3	
Sieve Size, inch	Percent Passing by Mass
1	---
3/4	---
1/2	100
3/8	90-100
No. 4	58-78
No. 8	40-60
No. 16	28-48
No. 30	18-38
No. 50	11-27
No. 100	6-18
No. 200	3-6

Job Mix Formula Requirements: Submit the proposed JMF in writing, for approval. Submittal must include the following:

- a. Percent passing each size.
- b. Optimum asphalt content.
- c. Percent of each aggregate and mineral filler to be used.
- d. Asphalt viscosity grade, penetration grade, or performance grade and additional test requirements as specified in paragraph ASPHALT BINDER.
- e. Number of blows of hammer per side of molded specimen. Number of Superpave gyratory compactor gyrations.
- f. Laboratory mixing and compaction temperatures.
- g. Supplier-recommended field mixing and compaction temperatures.
- h. Temperature-viscosity relationship of the asphalt binder.
- i. Plot of the combined gradation on the 0.45 power gradation chart, stating the nominal maximum size.
- j. Graphical plots and summary tabulation of stability, flow, air voids, voids in the mineral aggregate, and unit weight versus asphalt content as shown in AI MS-2. Include summary tabulation that includes individual specimen data for each specimen tested.
- k. Specific gravity and absorption of each aggregate.
- l. Percent natural sand.
- m. Percent particles with two or more fractured faces (in coarse aggregate).
- n. Fine aggregate angularity.
- o. Percent flat or elongated particles (in coarse aggregate).
- p. Tensile Strength Ratio and wet/dry specimen test results.
- q. Type and amount of antistrip agent (if required).

- r. List of all modifiers.
- s. Date the JMF was developed. Mix designs that are not dated or which are from a prior construction season may not be accepted.
- t. Warm-mix additive.